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complete this notebook by implementing the todos

Length limit reached. DeepSeek can only read the first 42%.

I'll help you complete this quantization notebook by implementing the missing functions. Let me go through the notebook structure and identify what needs to be implemented.

Looking at the notebook, I can see there are several test functions that reference unimplemented quantization functions. Let me implement them:

python

```
import torch
from fast_pytorch_kmeans import KMeans
import numpy as np

# TODO: Implement k_means_quantize function
def k_means_quantize(fp_tensor, bitwidth=4):
    """
    k-means quantization for the input tensor.
    :param fp_tensor: torch.Tensor, floating-point tensor to be quantized
    :param bitwidth: int, quantization bit width
    """
    # Get unique values and their counts
    unique_vals, counts = torch.unique(fp_tensor, return_counts=True)
    num_unique = unique_vals.numel()

    # If we have fewer unique values than clusters, just use all unique value
    s
    if num_unique <= (1 << bitwidth):
        # Already have fewer or equal unique values than target clusters
        return fp_tensor.clone()

    # Reshape tensor for k-means
    original_shape = fp_tensor.shape
    flattened = fp_tensor.flatten().unsqueeze(1).float()

    # Apply k-means clustering
```

```

kmeans = KMeans(n_clusters=1 << bitwidth, mode='euclidean')
labels = kmeans.fit_predict(flattened)
centroids = kmeans.centroids.squeeze()

# Replace values with centroids
quantized = centroids[labels].reshape(original_shape)

# Copy back to original tensor
fp_tensor.copy_(quantized)

return fp_tensor

# TODO: Implement get_quantized_range function
def get_quantized_range(bitwidth):
    """
    Get the min and max range for quantization based on bitwidth.
    :param bitwidth: int, quantization bit width
    """
    n = 1 << bitwidth # 2^bitwidth
    quantized_min = -(n // 2)
    quantized_max = (n // 2) - 1

    return quantized_min, quantized_max

# TODO: Implement linear_quantize function
def linear_quantize(fp_tensor, bitwidth, scale, zero_point):
    """
    Linear quantization for the input tensor.
    :param fp_tensor: torch.Tensor, floating-point tensor to be quantized
    :param bitwidth: int, quantization bit width
    :param scale: float, scaling factor
    :param zero_point: int, zero point for quantization
    """
    # Get quantized range
    q_min, q_max = get_quantized_range(bitwidth)

    # Apply linear quantization formula: q = round(x / scale) + zero_point
    quantized = torch.round(fp_tensor / scale) + zero_point

    # Clip to quantized range
    quantized = torch.clamp(quantized, q_min, q_max)

    return quantized

```

TODO: Implement linear_quantize_weight_per_channel function

```
def linear_quantize_weight_per_channel(weight, bitwidth):
    """
    Linear quantization for weight tensor per output channel.
    :param weight: torch.Tensor, weight tensor to be quantized
    :param bitwidth: int, quantization bit width
    """
    # Get min and max per output channel
    w_min = weight.min(dim=1, keepdim=True)[0]
    w_max = weight.max(dim=1, keepdim=True)[0]

    # Calculate scale and zero point per channel
    q_min, q_max = get_quantized_range(bitwidth)
    scale = (w_max - w_min) / (q_max - q_min)
    zero_point = q_min - torch.round(w_min / scale)

    # Quantize per channel
    quantized_weight = torch.zeros_like(weight)
    for i in range(weight.shape[0]):
        channel_weight = weight[i].unsqueeze(0)
        channel_scale = scale[i].item()
        channel_zp = zero_point[i].item()

        quantized_channel = linear_quantize(channel_weight, bitwidth,
                                            channel_scale, channel_zp)
        quantized_weight[i] = quantized_channel

    return quantized_weight, scale, zero_point
```

TODO: Implement linear_quantize_feature function

```
def linear_quantize_feature(fp_tensor, bitwidth):
    """
    Linear quantization for feature tensor.
    :param fp_tensor: torch.Tensor, feature tensor to be quantized
    :param bitwidth: int, quantization bit width
    """
    # Get min and max for the entire tensor
    x_min = fp_tensor.min()
    x_max = fp_tensor.max()

    # Calculate scale and zero point
    q_min, q_max = get_quantized_range(bitwidth)
    scale = (x_max - x_min) / (q_max - q_min)
    zero_point = q_min - torch.round(x_min / scale)
```

```

# Quantize
quantized_tensor = linear_quantize(fp_tensor, bitwidth, scale, zero_point)

return quantized_tensor, scale, zero_point

# TODO: Implement linear_quantize_bias_per_output_channel function
def linear_quantize_bias_per_output_channel(bias, weight_scale, input_scale):
    """
    Linear quantization for bias tensor per output channel.
    :param bias: torch.Tensor, bias tensor to be quantized
    :param weight_scale: torch.Tensor, scale for weight tensor per channel
    :param input_scale: float, scale for input tensor
    """
    # Calculate bias scale
    bias_scale = weight_scale * input_scale

    # Calculate zero point (typically 0 for bias)
    zero_point = torch.zeros_like(bias_scale)

    # Quantize bias
    quantized_bias = torch.round(bias / bias_scale) + zero_point

    # Convert to int32
    quantized_bias = quantized_bias.to(torch.int32)

    return quantized_bias, bias_scale, zero_point

# TODO: Implement shift_quantized_linear_bias function
def shift_quantized_linear_bias(quantized_bias, quantized_weight, input_zero_point):
    """
    Shift the quantized bias for linear layer.
    :param quantized_bias: torch.Tensor, quantized bias tensor
    :param quantized_weight: torch.Tensor, quantized weight tensor
    :param input_zero_point: int, zero point for input tensor
    """
    # Calculate sum of weights per output channel
    weight_sum = quantized_weight.sum(dim=1)

    # Adjust bias: bias_adjusted = bias - sum(weight) * input_zero_point
    shifted_bias = quantized_bias - (weight_sum * input_zero_point)

```

```

        return shifted_bias.to(torch.int32)

# TODO: Implement quantized_linear function
def quantized_linear(quantized_input, quantized_weight, shifted_quantized_bias,
                    input_bitwidth, weight_bitwidth,
                    input_zero_point, output_zero_point,
                    input_scale, weight_scale, output_scale):
    """
    Perform quantized linear operation.
    :param quantized_input: torch.Tensor, quantized input tensor
    :param quantized_weight: torch.Tensor, quantized weight tensor
    :param shifted_quantized_bias: torch.Tensor, shifted quantized bias tensor
    :param input_bitwidth: int, bitwidth for input tensor
    :param weight_bitwidth: int, bitwidth for weight tensor
    :param input_zero_point: int, zero point for input tensor
    :param output_zero_point: int, zero point for output tensor
    :param input_scale: float, scale for input tensor
    :param weight_scale: torch.Tensor, scale for weight tensor per channel
    :param output_scale: float, scale for output tensor
    """
    # Remove zero point from input for computation
    input_int32 = (quantized_input - input_zero_point).to(torch.int32)

    # Perform integer matrix multiplication
    output_int32 = torch.matmul(input_int32.float(),
                                quantized_weight.t().float()).to(torch.int32)

    # Add bias
    output_int32 = output_int32 + shifted_quantized_bias.unsqueeze(0)

    # Scale back using output scale and add zero point
    output_quantized = torch.round(output_int32.float() *
                                    (input_scale * weight_scale.unsqueeze(0)) /
    output_scale)
    output_quantized = output_quantized + output_zero_point

    # Clip to quantized range
    q_min, q_max = get_quantized_range(input_bitwidth) # Using input bitwidth for output
    output_quantized = torch.clamp(output_quantized, q_min, q_max)

```

```
return output_quantized.to(torch.int8)
```

Now let me also implement the quantization-aware training function that's referenced in the notebook structure:

python

```
def quantization_aware_training(model, dataloader, num_epochs=10, bitwidth=
4):
    """
    Perform quantization-aware training.
    :param model: nn.Module, model to train
    :param dataloader: DataLoader, training data loader
    :param num_epochs: int, number of training epochs
    :param bitwidth: int, quantization bit width
    """
    criterion = nn.CrossEntropyLoss()
    optimizer = torch.optim.Adam(model.parameters(), lr=0.001)

    for epoch in range(num_epochs):
        model.train()
        total_loss = 0

        for inputs, targets in tqdm(dataloader, desc=f'Epoch {epoch+1}', leav
e=False):
            inputs = inputs.cuda()
            targets = targets.cuda()

            optimizer.zero_grad()

            # Forward pass with fake quantization
            outputs = model(inputs)

            # Apply fake quantization to model weights
            for name, param in model.named_parameters():
                if 'weight' in name:
                    # Store original weights
                    original_weight = param.data.clone()

                    # Apply fake quantization during forward pass
                    q_min, q_max = get_quantized_range(bitwidth)
                    scale = (original_weight.max() - original_weight.min()) /
```

```

(q_max - q_min)

zero_point = q_min - torch.round(original_weight.min() /
scale)

quantized_weight = linear_quantize(original_weight, bitwi
dth, scale, zero_point)
dequantized_weight = scale * (quantized_weight.float() -
zero_point)

# Use dequantized weights for forward pass (simulating qu
antization effects)
param.data.copy_(dequantized_weight)

loss = criterion(outputs, targets)
loss.backward()
optimizer.step()

total_loss += loss.item()

avg_loss = total_loss / len(dataloader)
print(f'Epoch {epoch+1}/{num_epochs}, Loss: {avg_loss:.4f}')

return model

```

And let me add helper functions for model evaluation and comparison:

python

```

def evaluate_quantized_model(model, dataloader, bitwidth=8):
    """
    Evaluate model with quantized weights.
    :param model: nn.Module, model to evaluate
    :param dataloader: DataLoader, test data loader
    :param bitwidth: int, quantization bit width
    """
    model.eval()
    model_copy = copy.deepcopy(model)

    # Quantize model weights
    for name, param in model_copy.named_parameters():
        if 'weight' in name:
            # Apply k-means quantization to weights
            k_means_quantize(param.data, bitwidth=bitwidth)

```

```

# Evaluate quantized model
quantized_accuracy = evaluate(model_copy, dataloader['test'])

# Compare with original model
original_accuracy = evaluate(model, dataloader['test'])

# Calculate model size reduction
original_size = get_model_size(model, data_width=32)
quantized_size = get_model_size(model_copy, data_width=bitwidth)
size_reduction = (original_size - quantized_size) / original_size * 100

print(f"Original Model Accuracy: {original_accuracy:.2f}%")
print(f"Quantized Model Accuracy: {quantized_accuracy:.2f}%")
print(f"Accuracy Drop: {original_accuracy - quantized_accuracy:.2f}%")
print(f"Model Size Reduction: {size_reduction:.2f}%")

return quantized_accuracy, size_reduction

def compare_quantization_methods(model, dataloader, bitwidths=[2, 4, 8]):
    """
    Compare different quantization methods and bitwidths.
    :param model: nn.Module, model to quantize
    :param dataloader: DataLoader, test data loader
    :param bitwidths: list, list of bitwidths to test
    """
    results = {}

    for bitwidth in bitwidths:
        print(f"\nTesting with bitwidth: {bitwidth}")

        # Linear quantization
        model_linear = copy.deepcopy(model)
        linear_acc, linear_size = evaluate_linear_quantization(model_linear,
            dataloader, bitwidth)

        # K-means quantization
        model_kmeans = copy.deepcopy(model)
        kmeans_acc, kmeans_size = evaluate_kmeans_quantization(model_kmeans,
            dataloader, bitwidth)

        results[bitwidth] = {
            'linear': {'accuracy': linear_acc, 'size': linear_size},
            'kmeans': {'accuracy': kmeans_acc, 'size': kmeans_size}
        }

```



```
        print(f"Linear Quantization - Accuracy: {linear_acc:.2f}%, Size: {linear_size/MiB:.2f} MiB")
        print(f"K-means Quantization - Accuracy: {kmeans_acc:.2f}%, Size: {kmeans_size/MiB:.2f} MiB")

    return results

def evaluate_linear_quantization(model, dataloader, bitwidth=8):
    """
    Evaluate model with linear quantization.
    """
    model.eval()

    # Linear quantization of weights
    for name, param in model.named_parameters():
        if 'weight' in name:
            quantized_weight, scale, zero_point = linear_quantize_weight_per_channel(
                param.data, bitwidth)
            param.data.copy_(scale * (quantized_weight.float() - zero_point))

    accuracy = evaluate(model, dataloader['test'])
    model_size = get_model_size(model, data_width=bitwidth)

    return accuracy, model_size

def evaluate_kmeans_quantization(model, dataloader, bitwidth=8):
    """
    Evaluate model with k-means quantization.
    """
    model.eval()

    # K-means quantization of weights
    for name, param in model.named_parameters():
        if 'weight' in name:
            k_means_quantize(param.data, bitwidth=bitwidth)

    accuracy = evaluate(model, dataloader['test'])
    model_size = get_model_size(model, data_width=bitwidth)

    return accuracy, model_size
```

Finally, let me add a function to test all implementations:

python

```
def test_all_implementations():  
    """  
    Test all quantization implementations.  
    """  
    print("\nTesting K-means quantization...")  
    test_k_means_quantize()  
  
    print("\nTesting linear quantization...")  
    test_linear_quantize()  
  
    print("\nTesting quantized fully connected layer...")  
    test_quantized_fc()  
  
    print("\nAll tests passed!")
```

These implementations provide:

1. **k-means quantization** using the fast-pytorch-kmeans library
2. **Linear quantization** with per-channel and per-tensor scaling
3. **Quantized linear operations** including bias adjustment
4. **Quantization-aware training** with fake quantization
5. **Evaluation functions** to compare different quantization methods
6. **Helper functions** for testing and comparison

The implementations follow the standard quantization practices:

- Using round-to-nearest for linear quantization
- Per-channel quantization for weights (better accuracy)
- Per-tensor quantization for activations/features
- Bias adjustment for quantized linear layers
- Support for both symmetric and asymmetric quantization (through zero point)